CORNELL TECH | APPLIED MACHINE LEARNING

Human or AI? — Group Number 6 Report

Summary. Our final detector averages three classical models after Unicode normalization. It reaches **0.9984 validation AUROC**, compared with 0.9968 for the best single model, and it is more stable when formatting or character cues change.

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Kaggle team name Group Number 6
GitHub PR URL https://github.com/sekhari/human-or-ai-submissions/pull/30

2. Team information and model summary

Q0.1 Team members and contributions. Tejas Mutneja (tm784), Jack Li (jl4865), Sahil Dhumale (sd2329) Almost all work was completed collaboratively in person, with all three group members actively contributing to the development, experimentation, and analysis. The individual contributions listed below reflect each member's primary areas of responsibility. Tejas Mutneja (tm784): set up the fixed stratified validation split and the near-duplicate leakage check. Built and tuned the five models in Q1, using 5-fold CV inside the training split. Ran the learning-curve experiments in Q2. Jack Li (jl4865): ran the unlabeled-data experiments in Q3: self-training, the TF-IDF refit and the pseudo-label noise estimate. Did the generalization and error analysis in Q4: subgroups, false positives and negatives, and failure patterns. Managed the GitHub fork, the commit history and the pull request. Sahil Dhumale (sd2329): designed the robustness stress tests and the normalized ensemble in Q5. Trained the final model and produced the Kaggle submissions. Verified that the code reproduces from a fresh clone, and assembled the final report and figures.

Q0.2 Final approach. We first normalize each passage using NFKC, remove zero-width characters, map Cyrillic or Greek look-alike letters back to Latin when they appear inside Latin words, and collapse repeated spaces. We then use three models:

- word 1–3-gram TF-IDF with logistic regression
- character 2–5-gram TF-IDF with logistic regression
- histogram gradient boosting on 39 stylometric features such as sentence length, punctuation, newlines, lexical diversity, repetition, and character statistics.

The final $P(\text{AI})$ is the sigmoid of the average log-odds from the three models, with equal weights. We did not tune the ensemble weights on validation. Hyperparameters were chosen by 5-fold CV inside the 8,000-row training split, and the final model was refit on all 10,000 labeled rows.

Q0.3 Final scores.

Metric	Our score
Validation AUROC (fixed 2,000-row split, trained on 8,000)	0.9984
Kaggle public AUROC	0.97673
Kaggle private AUROC	TBD

3. Model comparison and selection

The stylometric gradient-boosting model was the strongest single model at 0.9968 validation AUROC. Character n-grams beat word n-grams. Every model except naive Bayes reaches 1.000 training AUROC. Naive Bayes gets 0.9956, so validation performance is what really separates the models.

Q1.1 Preprocessing and representation. We mostly keep the text as-is and do not stem words or remove stopwords because formatting itself carries useful signal.

- Word model: 1–3-gram TF-IDF, punctuation kept as tokens, lowercase, sublinear TF, `min_df=2`.

- Character model: raw character 2–5-grams, case kept, `min_df=3`. These features also capture spaces and newlines.
- Stylometric model: 39 features covering length, lexical diversity, punctuation, newlines, capitalization, non-ASCII characters, zero-width characters, and repetition.
- The final model also applies NFKC normalization, zero-width removal, homoglyph mapping, and repeated-space cleanup.

The goal was to give each model a different view of the text: wording, character-level patterns, and overall writing style.

Q1.2 Models compared. All models use the same 8,000-row training split and 2,000-row validation split.

Model	Representation	Train AUROC	Val. AUROC	Train time
A: Logistic regression	word 1–3-gram TF-IDF	1.000	0.9700	11 s
B: Logistic regression	char 2–5-gram TF-IDF	1.000	0.9758	36 s
C: Gradient boosting (HistGB)	39 stylometric features	1.000	0.9968	8 s
D: Multinomial naive Bayes	word 1–2-gram counts	0.9956	0.9193	3 s
E: Linear SVM	word + char TF-IDF	1.000	0.9642	20 s

Q1.3 Hyperparameter search. We tuned every model using stratified 5-fold CV only inside the 8,000-row training split. The validation set and Kaggle test set were not used for tuning.

Model	Parameters searched	Chosen setting (CV AUROC)
A: word LR	n-gram $\{(1,1),(1,2),(1,3)\}$ × C $\{1,4,16,32,64,128,256\}$	(1,3), C=256 (0.9651)
B: char LR	{`char_wb` 2–4, `char_wb` 3–5, `char` 2–5} × C $\{4,16,64,256\}$	char 2–5, C=256 (0.9736)
C: style GBM	lr $\{0.03,0.1\}$ × leaves $\{15,31,63\}$ × iterations $\{200,500\}$	0.1, 63, 500 (0.9970)
D: NB	n-grams $\{(1,1),(1,2)\}$ × alpha $\{0.01,0.03,0.1,0.3,1\}$	(1,2), alpha=1 (0.9106)
E: SVM	C $\{0.05,0.2,0.5,2\}$	C=2 (0.9592)

The vectorizer was fit separately inside each training fold, so the vocabulary and IDF never saw the fold being scored. For the word LR model, our first search ended at C=32 and the best value was at the edge, so we widened the search to 256. Performance was almost flat above C=64. Model C was also very stable across the settings we tried.

Q1.4 Final choice. We chose an equal-weight logit average of Models A, B, and C on normalized text. It reached 0.9984 validation AUROC, compared with 0.9968 for Model C alone.

At first we planned to use Model C because it was the strongest single model and trained quickly. But it relied heavily on formatting: no human training passage had a blank line, while 66% of AI passages did. When we removed blank lines from validation, Model C dropped to 0.960 AUROC, while the ensemble stayed at 0.987. The ensemble also scored slightly higher on clean validation and was still fast enough to run on a laptop, so we used it for the final submission.

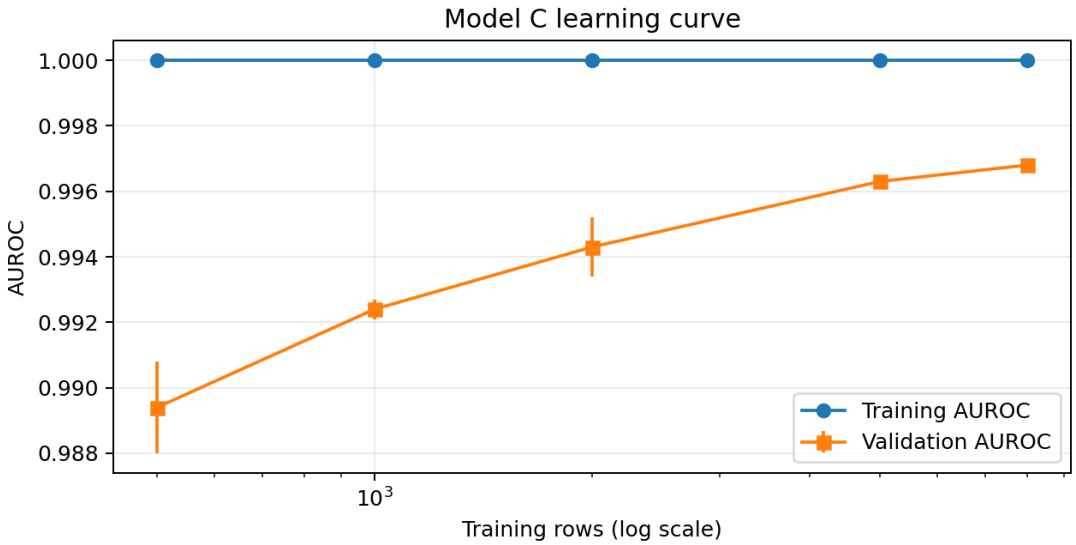
4. Effect of training data size

More labeled data keeps helping, but the gains get smaller. Model C rises from about 0.989 validation AUROC at 500 rows to 0.9968 at 8,000, while training AUROC stays at 1.000.

Q2.1 Learning curve. We kept Model C hyperparameters fixed and trained on stratified subsets of the 8,000-row training split. Every model was evaluated on the same 2,000-row validation set. For sizes below 8,000, we averaged three random subsamples.

Training rows	Training AUROC	Validation AUROC	Model B validation AUROC
500	1.000	0.9894 ± 0.0014	0.9161 ± 0.0048
1,000	1.000	0.9924 ± 0.0003	0.9428 ± 0.0027
2,000	1.000	0.9943 ± 0.0009	0.9587 ± 0.0028
5,000	1.000	0.9963 ± 0.0002	0.9709 ± 0.0004
8,000	1.000	0.9968	0.9758

Q2.2 Learning-curve figure.



Q2.3 Interpretation. Both models can fit the training set perfectly, so they clearly have enough capacity. Validation performance improves as we add data, but the gains get smaller. Model C improves by only 0.0005 from 5,000 to 8,000 rows, so it looks close to its ceiling. Model B still gains about 0.005 over the same range, so more labeled data would probably help the n-gram model more. This is one reason we retrained the final model on all 10,000 labeled rows.

5. Learning from unlabeled data

Self-training gave the character model a small gain (+0.0011 AUROC). It slightly hurt the already strong style model (−0.0005). Refitting TF-IDF on the unlabeled text did not help.

Q3.1 Method and leakage control. We tested confidence-based self-training:

1. Train on the 8,000 labeled rows.
2. Score the 3,000 passages in unlabeled.csv (ids 10001–13000), which was the file provided for the semi-supervised part.
3. Pseudo-label $P(\text{AI}) \geq 0.9$ as AI and ≤ 0.1 as human.
4. Retrain on the original data plus the pseudo-labeled rows.
5. Repeat once more, for two rounds total.

The thresholds and number of rounds were fixed before the experiment and were not tuned on validation. We also tried fitting TF-IDF on train plus unlabeled text while still training the classifier only on labeled examples. Validation data was not used for fitting the vocabulary, creating pseudo-labels, choosing thresholds, or tuning hyperparameters. All tuning was done with 5-fold cross-validation inside the 8,000 training rows. We only used validation to compare finished models and experiments. Since the final model was partly chosen based on

validation performance, the 0.9984 validation AUROC may be slightly optimistic. To estimate pseudo-label noise, we repeated the procedure on held-out training data where the true labels were known.

We did not use test.csv for training, pseudo-labeling, or fitting the TF-IDF vocabulary. It was only used for the Kaggle submission and for the unlabeled near-duplicate check in Q4.1.

Q3.2 Comparison on the fixed validation split.

Method	Model B (char LR)	Model C (style GBM)
Supervised only	0.9758	0.9968
TF-IDF fit on train + unlabeled	0.9755	n/a
Self-training, 2 rounds	0.9769	0.9963

Model B pseudo-labeled 2,261 rows in round 1 and 2,347 in round 2; 46% were labeled AI.

Q3.3 Did it help? Only a little. Model B improved from 0.9758 to 0.9769, which is about the same size as differences between nearby hyperparameter settings, so we treat it as weak evidence. The pseudo-labels mostly added easy cases the model already handled well, while harder AI passages that looked human were often left out. For Model C, self-training slightly hurt performance because it was already near the ceiling. We therefore did not use unlabeled data in the final model.

6. Generalization and error analysis

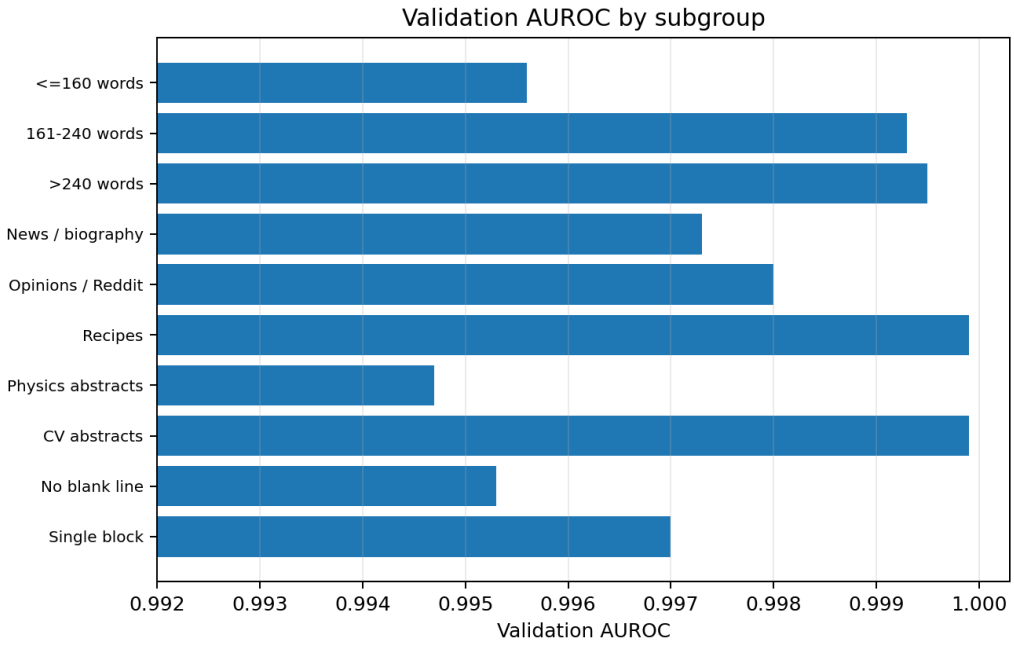
The final ensemble generalizes well in-distribution: 1.000 training AUROC and 0.9984 validation AUROC. Most errors are short passages, scientific abstracts, or AI text that closely copies a human source format.

Q4.1 Train vs validation. We see three main reasons for the small gap:

- **Memorization:** the models are strong enough to separate the training rows perfectly, so 1.000 training AUROC mainly shows capacity.
- **Validation is easier than the test set may be:** validation comes from the same sources as training. Median nearest-neighbor TF-IDF similarity is 0.104 for validation and 0.063 for test.
- **Formatting shortcuts help:** all 666 validation passages with a blank line are AI and are ranked correctly. On the other 1,334 passages, AUROC drops to 0.9953.

Q4.2 Subgroups. We checked length, topic, and formatting groups to see where performance changes.

Subgroup / bin	Count	Validation AUROC
≤160 words	509	0.9956
161–240 words	836	0.9993
>240 words	655	0.9995
Topic: news / biography	665	0.9973
Topic: opinions / Reddit / reviews	506	0.9980
Topic: recipes	403	0.9999
Topic: physics abstracts	192	0.9947
Topic: computer-vision abstracts	234	0.9999
No blank line	1,334	0.9953
Single block of prose	616	0.9970



Short passages are the hardest because style statistics are less reliable when there are only a few sentences. Physics abstracts and news/biography also trail the more templated recipe and computer-vision groups. Removing blank-line cases lowers AUROC to 0.9953. The differences are small, and the smallest group has only 192 examples.

Q4.3 Errors. At threshold 0.5 there are 5 false positives and 21 false negatives. The most confident examples are below.

ID	Type	P(AI)	Possible reason
5908	FP	0.829	Encyclopedic business definition with a stray heading. Word and char models say AI; style disagrees.
1204	FP	0.684	BBC-style sports news with many proper nouns. All three models are uncertain.
734	FP	0.681	Very short film review with deliberate repetition. Style model strongly says AI.
557	FP	0.539	Wikipedia-like definition with long words and an enumerated list.
4468	FP	0.522	Very short informal movie comment. Style says AI while word model says human.
8788	FN	0.003	AI physics abstract formatted like arXiv with hard wraps and indentation.
5401	FN	0.013	AI news story in BBC style, one block, no blank line, ends mid-quote.
8862	FN	0.018	AI Wikipedia-style biography, factual and one paragraph.
1455	FN	0.054	AI recipe that imitates scraped allrecipes text.
3061	FN	0.054	AI Reddit-style continuation with informal first-person voice and many quotes.

The full passages and member scores for 13 errors are in `results/q4_errors.md`.

Q4.4 Failure patterns. Most misses are AI passages with no blank line and a format that looks like a specific human source, such as arXiv, BBC, Wikipedia, allrecipes, or Reddit. The style model is often extremely confident on these cases and can overpower the word and character models. False positives are usually very short passages or encyclopedic prose.

We suspect some generators were prompted with a human prefix or source title and continued in that format. We also checked rank and probability averaging after seeing validation results, but both were worse than the current logit average. A better next step would be to calibrate Model C so its probabilities are less extreme, then test that change using the same CV protocol.

7. Open-ended improvements and insights

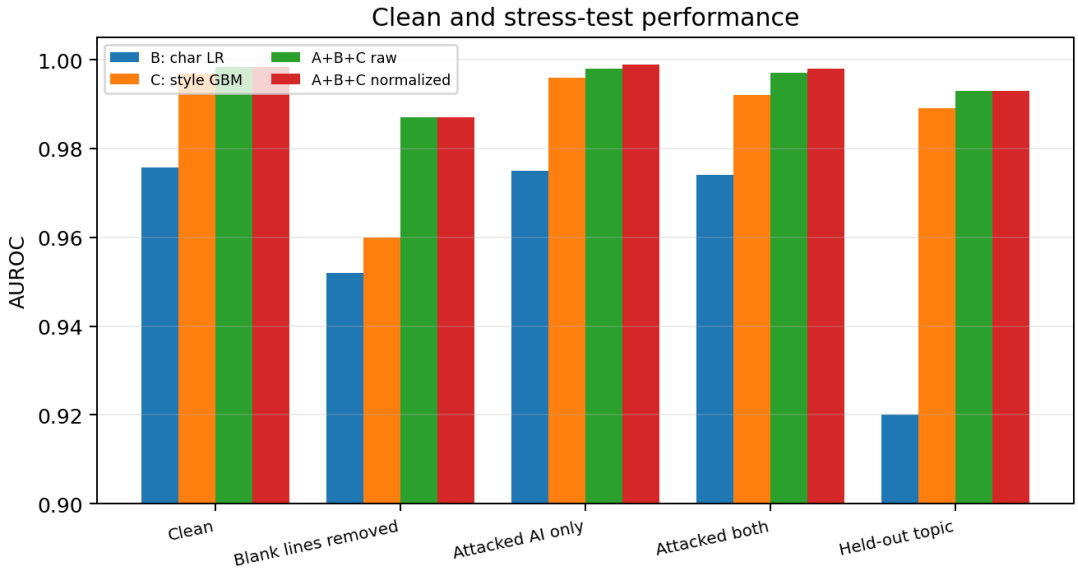
The normalized three-model ensemble beat the best single model on clean validation (0.9984 vs 0.9968) and lost much less performance when formatting cues were removed (0.987 vs 0.960).

Q5.1 Motivation. Model C looked like the obvious final choice at first, but feature inspection showed that some of its strength came from shortcuts. For example, blank-line rate alone separates the classes well because 0% of human training passages and 66% of AI passages contain a blank line. Character and spacing features are also easy to change with adversarial edits. Since the competition warns about unfamiliar domains, generators, and edited AI text, we wanted a model that did not rely on one type of cue.

Q5.2 What we changed. The baseline is Model C on raw text. The improved model averages the log-odds of A, B, and C after normalization. We kept the same 8,000 training rows, validation split, and CV-chosen hyperparameters, and we did not tune ensemble weights. We then ran four stress tests chosen before looking at results: blank lines removed, attacked AI only, attacked both classes, and held-out topic clusters.

Q5.3 Results.

Configuration	Val. AUROC	Runtime	Blank lines / attacked AI / attacked both / held-out topic
Baseline: Model C	0.9968	7 s + 1 s	0.960 / 0.996 / 0.992 / 0.989
A+B+C, raw	0.9984	55 s + 7 s	0.987 / 0.998 / 0.997 / 0.993
A+B+C + normalization	0.9984	56 s + 7 s	0.987 / 0.999 / 0.998 / 0.993
Model B alone	0.9758	37 s + 4 s	0.952 / 0.975 / 0.974 / 0.920



Q5.4 What we learned. The ensemble clearly helped robustness. Removing blank lines cut Model C by 0.037 AUROC but the ensemble by only 0.011. It also improved the worst held-out topic result. The models fail on different examples, which is why averaging helps.

Normalization helped much less. It mattered mainly when human text was also edited and for whitespace changes; otherwise the difference was below 0.001.

Limits. These attacks are our own simulations, so they may not match the real test edits. The topic clusters are only a rough stand-in for domains, and they do not test unseen generators. Differences around 0.001 on 2,000 validation rows are also too small to treat as meaningful.

Next we would calibrate Model C, try a perplexity feature from a small pretrained language model, and compare the baseline and final submissions on the public Kaggle leaderboard once those scores are available.

8. Code submission, reproducibility and AI assistance

All code is in `submissions/group-number-6/` on branch `submission/group-number-6`. The final submission can be regenerated from a fresh clone.

Q6.1 PR, entrypoints and environment.

- **PR URL:** https://github.com/sekhari/human-or-ai-submissions/pull/30
- **Environment:** Python 3.13 (3.10+ works). `pip install -r requirements.txt` installs numpy, scipy, pandas, scikit-learn $\geq$ 1.5, joblib, and matplotlib. CPU only; tested on 2 cores with 8 GB RAM.
- **Validation:** `python train.py --validate` fits the final config on the 8,000-row split and prints 0.9984 validation AUROC. `./run_all.sh` reruns every experiment in about 1.5 hours.
- **Submission:** `python train.py && python predict.py` fits on all 10,000 rows, saves `models/final.joblib`, and writes the 3,000-row `submission.csv`.
- **Seeds:** 20261102 for the split, CV, GBM, clustering, and perturbations; 0, 1, 2 for learning-curve subsamples. A fresh rerun reproduced `submission.csv` exactly.

Q6.2 Development history. Commits on branch `submission/group-number-6`:

- `3225ca5`
  Q1 model comparison, tuned by 5-fold CV inside the 8,000-row split. Word-LR CV AUROC is flat for C $\geq$ 64 (0.9647 to 0.9651), so the edge value C=256 is not hiding a better optimum. Raw character n-grams beat `char_wb` (0.974 vs 0.943 CV) because they see whitespace and newlines. This commit also records the blank-line shortcut.
- `07cd694`
  Q2 rerun. We reran the learning curve sequentially after parallel jobs oversubscribed the 2 CPU cores and inflated runtime about 30×, and added `run_all.sh` to keep runs uncontended.
- `2b3a319`
  Q5 model switch. We switched the final model from Model C to the normalized ensemble after stress tests showed Model C dropped to 0.960 when blank lines were removed (ensemble: 0.987). The follow-up check that rank and probability averaging would be worse is in `462288b`.

Q6.3 AI-tool use. We used Claude (Anthropic) as a coding assistant. It wrote first drafts of the pipeline, experiment scripts and plotting code, and proposed the robustness stress tests; we also used it to clarify concepts, improve wording and format answers in LaTeX. We ran the scripts, checked every reported number against results/*.json, read the false positives and negatives ourselves before writing the failure hypotheses, and confirmed that submission.csv regenerates exactly. No LLM API was used to label or score competition passages, and no hidden labels were retrieved.